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Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System

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Statement of Authorship

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Abstract

In-cabin occupant monitoring is increasingly required by vehicle safety regulation and by advanced restraint systems that must adapt to who is seated where. This thesis investigates the fusion of RGB camera images with pressure heatmaps obtained from seat-embedded pressure sensors in order to jointly estimate seat occupancy, passenger class, occupant weight, and body pose. A synthetic pressure heatmap generation pipeline based on human body simulation is proposed to complement limited real sensor data, together with a Sim2Real validation procedure that quantifies the remaining domain gap. A multi-task deep learning fusion architecture combining camera and pressure branches is designed, trained, and evaluated against single-modality baselines through ablation studies. The expected contribution is a validated fusion pipeline and dataset methodology that improves robustness of occupant state estimation over either sensing modality used in isolation.

Keywords: cabin monitoring, sensor fusion, pressure heatmap, occupant classification, pose estimation, Sim2Real, multi-task learning

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List of Abbreviations

Abbreviation	Meaning
CNN	Convolutional Neural Network
DMS	Driver Monitoring System
EKF	Extended Kalman Filter
FOV	Field of View
GDPR	General Data Protection Regulation
IMU	Inertial Measurement Unit
IR	Infrared
MSE	Mean Squared Error
OMS	Occupant Monitoring System
RGB-D	Red-Green-Blue plus Depth (image with depth channel)
ROI	Region of Interest
Sim2Real	Simulation-to-Reality (transfer)
ToF	Time-of-Flight
UKF	Unscented Kalman Filter
WRMSE	Weighted Root Mean Squared Error

1 Introduction

1.1 Motivation and Relevance

Occupant monitoring is becoming more important as vehicle interiors and driving roles evolve. Highly automated vehicles may allow occupants to adopt reclined, rotated, or otherwise non-nominal seating postures that conventional restraint systems were not primarily designed around [1]. Increasing automation is also associated with a broader range of non-driving-related activities, which can change occupant posture and driver readiness [2]

These changes are safety-relevant because occupant posture strongly influences restraint interaction and injury risk. Real-world crash analysis using GIDAS data showed that belted reclined occupants had higher odds of MAIS,2+, MAIS,3+, and MAIS,4+ injury than occupants seated upright [3]. In conditionally automated driving, non-driving postures can also increase takeover time and reduce takeover quality [2].

Accurate occupant information is also important for airbag control. Although frontal airbags have saved more than 50,000 lives, historical NHTSA investigations documented airbag-induced fatalities, particularly among children and occupants positioned close to the airbag module [4, 5]. Such cases motivated advanced restraint systems capable of suppressing or adapting airbag deployment based on occupant characteristics and seating conditions [6]. Information such as occupant presence, weight, seated posture, and classification can therefore influence safety-critical restraint decisions.

Regulation and consumer-safety assessment increasingly reflect this need. FMVSS No. 208 introduced advanced-airbag performance requirements intended to reduce airbag-related injury to children and small occupants while maintaining frontal-crash protection . UN Regulation No. 157 requires automated driving systems to assess driver availability before a transition of control [7], while European regulations also require advanced driver-distraction warning functions [8]. Euro NCAP's 2026 protocols place greater emphasis on driver and occupant monitoring, seat-belt-use detection, occupant information, and restraint adaptation [9].

A single sensing modality, however, may not provide sufficiently robust occupant-state estimation under all cabin conditions. Camera-based monitoring can provide detailed information about body posture, appearance, and keypoints, but its performance can degrade under poor illumination, glare, shadows, or occlusion [10]. Seat-pressure sensing is independent of illumination and provides direct information about body-seat contact and loading, making it useful for occupant detection and classification [11]. However, pressure data alone may be ambiguous when different occupants, objects, or postures generate similar contact patterns.

Camera and pressure sensing are therefore complementary modalities. Vision provides semantic and geometric information, while pressure sensing provides physical contact and load information. Multimodal fusion can exploit this complementarity to improve robustness when one modality becomes uncertain or partially degraded [12]. The central objective of this thesis is therefore to determine whether fusing RGB images and seat-pressure distribution maps improves the estimation of occupancy, occupant class, posture, and weight compared with unimodal models.

1.2 Problem Statement

Reliable occupant monitoring requires the estimation of multiple occupant states, including seat occupancy, occupant class, posture, and weight or weight category. This requirement is becoming increasingly relevant as current vehicle-safety assessment frameworks place greater emphasis on occupant monitoring and the adaptation of safety functions to occupant characteristics [9]. However, obtaining these states reliably from a single sensing modality remains challenging.

Camera-based systems provide rich semantic and geometric information for detecting occupants and estimating body posture, but their performance can deteriorate under poor illumination, glare, shadows, and partial occlusion [10]. Seat-pressure sensing avoids these visual limitations and can capture body-seat contact, loading, and posture-related pressure distributions. Even relatively low-resolution pressure configurations have demonstrated useful posture-recognition capability [13]. Nevertheless, recent pressure-based studies show that performance can degrade when seating conditions change and that geometrically similar postures may produce ambiguous pressure patterns [14]. Thus, the two sensing modalities provide complementary information but also exhibit different failure modes.

Multimodal in-cabin monitoring has already demonstrated that heterogeneous signals such as video, seat pressure, and physiological measurements can be combined successfully for specific tasks such as driver takeover-state recognition [15, 16]. However, within the literature considered in this thesis, the benefit of jointly using camera images and seat-pressure maps for a unified set of occupant-monitoring functions —particularly occupancy, occupant classification, posture, and weight estimation—has not been sufficiently quantified against equivalent unimodal baselines.

The central problem addressed in this thesis is therefore to determine whether learned fusion of camera and seat-pressure information provides a measurable improvement in occupant-state estimation compared with camera-only and pressure-only models. This requires developing modality-specific representations, synchronizing the heterogeneous sensor streams, designing an appropriate fusion strategy, and evaluating the fused model under the same data and performance criteria as the corresponding unimodal systems.

1.3 Research Questions

"To what extent does feature-level fusion of synchronized RGB cabin images and low-resolution seat-pressure distribution maps improve the classification of seat occupancy, seated posture, and occupant size class compared with equivalent RGB-only and pressure-only models."

1. RQ1.1: How do the RGB-only and pressure-only models compare in classification performance across seat occupancy, seated posture, and occupant weight classification?

2. RQ1.2: To what extent does the fused RGB–pressure model improve classification performance over the best-performing unimodal baseline for each classification task?

3. RQ1.3: What contribution do sensor weighted combination(WC) and cross-sensor correlation(CS) fusion framework does to the classification performance of the RGB–pressure fusion model?

4. RQ1.4: How robust is RGB–pressure fusion to controlled degradation of either sensing modality compared with the corresponding unimodal models?

Terminology: "modal/modality" refers to a distinct source of sensory information. RGB images constitute the visual modality, seat-pressure distribution maps constitute the pressure modality. "model" refers to the learned computational architecture used to process one or both modalities.

Occupant state = the overall concept being estimated. Occupant-state attributes = occupancy, posture, and size/weight class. Classes = the discrete labels within each attribute.

1.4 Scope and Delimitations

The scope of this thesis is restricted to supervised, per-seat occupant-state attributes estimation using synchronized RGB image and low-resolution seat-pressure distribution maps. The considered occupant-state attributes are occupancy, occupant class, seated posture, and occupant weight or weight category. Separate RGB and pressure encoders are used to establish unimodal baselines, and their learned representations are combined using a feature-level heterogeneous fusion . [16]. The study evaluates whether fusion improves predictive performance relative to the individual modalities under an identical dataset, data split, and evaluation protocol.

This thesis investigates supervised, per-seat occupant-state attributes using two heterogeneous sensing modalities: RGB cabin images and low-resolution seat-pressure distribution maps. The study is deliberately restricted to adult occupants and considers three classification attributes: seat occupancy, seated posture, and occupant weight class.

Seat occupancy is formulated as a binary classification problem with the classes *occupied* and *empty*.

For adult-occupied seats, the seated-posture attribute is formulated as a four-class classification problem comprising *upright*, *forward lean*, *reclined*, and *side lean*. These classes are selected to represent a nominal seated posture and three common non-nominal deviations that are relevant to occupant monitoring and restraint interaction. Postures outside these predefined categories are not considered in this study.

The occupant-weight attribute is formulated as a three-class classification problem comprising *light*, *medium*, and *heavy* adult occupants. The considered body-mass range is restricted to approximately 49–101.2 kg and is selected to span representative small, mid-size, and large adult occupants commonly used in automotive safety evaluation. The classes are defined as *light* (49–< 63.35 kg), *medium* (63.35–< 89.45 kg), and *heavy* (89.45–101.2 kg). These boundaries are study-specific classification thresholds; children and occupants outside the defined adult range are not considered.

The study considers two sensing modalities: RGB cabin images and synthetic low-resolution seat-pressure distribution maps. The pressure maps are generated through physics-based simulation and are designed to reproduce the spatial structure and resolution of the target pressure-sensor configuration rather than to model the exact physical pressure values measured by a real sensor.

The conclusions of this thesis are therefore limited to classification performance using the simulated pressure representation. Transfer of the trained pressure or fusion models to a physical pressure mat, and quantitative validation of the simulated pressure distributions against real measurements, are outside the scope of this work.

RGB-only and pressure-only models are evaluated as unimodal baselines and compared with a feature-level heterogeneous fusion model adapted from DeepFusion [16]. The study additionally evaluates the contribution of the Weighted-Combination and Cross-Sensor components through ablation experiments and examines model robustness under controlled degradation of either modality.

The objective is to evaluate the effectiveness and robustness of RGB–pressure fusion for the defined occupant-state classification tasks rather than to develop or validate a complete production-ready occupant monitoring system.

Our sensor's: Capacitative Pressure sensor mat : Low resultion sensor, presented as a 2D heatmap,

RGB camera: 2D image sensor, presensted as a 2D image,

These two sensors are heterogeneous, due to their different sensing modalities, data formats, and information content.

Fusion Approaches:

1. Signal / data-level (early fusion) — combine raw or lightly-processed measurements before feature extraction. Concatenating raw CSI with raw accelerometer streams; Kalman-filtering redundant range sensors.

Best when: modalities are homogeneous, synchronized.

2. Feature-level (mid-level fusion) — combine learned feature representations from each modality.

Advantages: handles heterogeneity naturally; dimensionality reduction before fusion; still learns joint cross-modal structure; robust to per-sensor scale differences. Best : when modalities are comepletely heterogeneous, or modalities differ but jointly informative about same aspect .

3.Decision-level (late fusion): each sensor gets its own full classifier; outputs combined by voting, averaging, Bayesian combination etc

Advantages: maximum modularity — sensors can be added, removed, or fail gracefully; each branch trains independently, sometimes on differently-sized datasets;

Costs: discards all cross-modal correlation — this is the big one. If a modality is only informative in combination with another, late fusion cannot see it. Also needs each modality to be individually sufficient for the task.

Deep fusion architectures can combine multiple fusion strategies: 1. Per-branch encoders + concat (SR+Avg-style) 2.Attention-weighted 3.correlation modeling 4.Cross-modal attention

Devices 2× smartphone, 1× smart watch, 1× Shimmer

1.5 Thesis Structure

Chapter 2 introduces the background and fundamentals of cabin monitoring systems, pressure sensing technology, camera systems, and sensor fusion theory required to follow the remainder of the thesis. Chapter 3 reviews related work on pressure-based and camera-based occupant monitoring, multi-modal fusion approaches, and synthetic data generation, concluding with a gap analysis that positions this work. Chapter 4 defines the use cases, functional and non-functional requirements, and overall system architecture. Chapter 5 describes the real data collection protocol, the synthetic pressure heatmap generation pipeline, and the Sim2Real validation procedure. Chapter 6 presents the preprocessing and temporal synchronisation pipeline that prepares camera and pressure data for model training. Chapter 7 details the multi-task fusion architecture, loss formulation, and training configuration. Chapter 8 reports quantitative results, ablation studies, and Sim2Real transfer evaluation. Finally, Chapter 9 summarises the contributions, answers the research questions, discusses limitations, and outlines future work.

2 Background and Fundamentals

2.1 Cabin Monitoring Systems

Cabin monitoring systems are commonly divided into driver monitoring systems (DMS), which focus on the driver's attentiveness and drowsiness, and occupant monitoring systems (OMS), which assess the presence, position, and classification of all vehicle occupants. Regulatory bodies such as Euro NCAP have progressively increased the weight given to occupant status monitoring in their safety rating protocols, driving adoption of both camera-based and pressure-based sensing across vehicle segments [17]. Typical OMS use cases include seatbelt reminder logic, smart airbag suppression for child seats, and post-crash occupant status reporting for emergency services.

2.2 Pressure Sensing Technology

2.2.1 Working Principle and Sensor Types

Seat-integrated pressure sensors typically use resistive, capacitive, or piezoelectric transduction principles arranged in a grid to produce a spatially resolved pressure map of the seat surface. Resistive force sensing resistor (FSR) arrays are low-cost and widely used in automotive seat occupancy classification, while capacitive mats offer higher sensitivity at increased manufacturing cost [18]. The choice of sensor technology directly affects spatial resolution, sampling rate, and susceptibility to drift over the vehicle's operating temperature range.

[19] This is paper on lower resolution pressure sensor array for driver posture monitoring and pose estimation.

2.2.2 Pressure Heatmap Representation

The raw output of a pressure sensor grid is commonly represented as a two-dimensional heatmap, where each pixel encodes the normalised pressure value at a corresponding physical location on the seat surface. This representation allows pressure data to be processed with the same convolutional architectures typically applied to single-channel images, which is a key enabler for the fusion architecture proposed in Chapter 7.

2.3 Camera Systems for In-Cabin Use

2.3.1 RGB and IR Cameras

Standard RGB cameras provide high-resolution colour imagery suitable for detailed pose and appearance analysis but perform poorly under low-light or backlit conditions common in vehicle cabins. Near-infrared (IR) cameras paired with active IR illumination maintain consistent image quality irrespective of ambient lighting and are therefore preferred in production driver and occupant monitoring systems [20].

2.3.2 Depth Cameras

2.4 Sensor Fusion Fundamentals

2.4.1 Fusion Levels

Sensor fusion approaches are typically categorised into data-level fusion (combining raw signals before any feature extraction), feature-level fusion (combining learned intermediate representations), and decision-level fusion (combining independent per-modality predictions). Feature-level fusion generally offers the best balance between representational flexibility and robustness to modality-specific noise, and is the strategy adopted for the architecture in this thesis [21].

2.4.2 Deep Learning-Based Fusion

Deep learning-based fusion replaces hand-crafted process and measurement models with learned convolutional or attention-based feature extractors per modality, followed by a learned fusion module, building on the general deep learning foundations of representation learning and gradient-based training described in [22]. This approach scales naturally to high-dimensional, spatially structured inputs such as camera images and pressure heatmaps, and can be trained end-to-end jointly with the downstream task heads [23].

2.5 Multi-Task Learning

Multi-task learning trains a single shared backbone with multiple task-specific output heads, allowing related tasks to regularise one another and share statistical strength, which is particularly valuable when labelled data for some tasks (e.g. weight estimation) is scarcer than for others (e.g. binary occupancy). A central challenge in multi-task learning is balancing the loss contributions of tasks with different scales and convergence rates, which is addressed explicitly for this thesis in Section 7.6.

2.6 Human Body Modelling for Simulation

Parametric human body models represent body shape and pose using a low-dimensional set of shape and joint-angle parameters, enabling generation of anatomically plausible body meshes across a wide population of statures and weights. Combined with a physics-based seat contact simulation, such models allow synthetic pressure heatmaps to be rendered for arbitrary occupant configurations without requiring a physical subject, which forms the basis of the synthetic data generation pipeline described in Section 5.4 [24].

3 Related Work and State of the Art

3.1 Pressure-Based Occupant Detection and Classification

Early automotive seat occupancy detection relied on simple threshold-based binary weight switches, which were later extended to spatially resolved pressure mats capable of distinguishing occupant classes beyond a simple occupied/unoccupied signal [18]. Subsequent work applied regression neural networks directly to pressure heatmaps to estimate continuous occupant weight, demonstrating that spatial pressure distribution carries substantially more information than aggregate load alone [25]. A recent survey consolidates these pressure-sensing approaches and highlights classification accuracy, sensor cost, and robustness to seat cover variation as the primary axes along which such systems are compared [26].

3.2 Camera-Based Cabin Monitoring

Camera-based approaches to cabin monitoring have progressed from frontal-face driver drowsiness detection toward full-cabin, multi-occupant pose estimation. Infrared camera systems combined with real-time pose estimation pipelines have been shown to operate reliably across day and night conditions inside the vehicle cabin [20]. Lightweight convolutional architectures tailored to the automotive embedded compute budget have been proposed specifically for occupant pose classification [27], while RGB-D fusion methods combine colour and depth cues to improve robustness against variable illumination and occluding objects such as blankets or bags placed on seats [28].

3.3 Multi-Modal Sensor Fusion Approaches

Beyond the automotive cabin monitoring domain, multimodal deep fusion has been studied extensively for tasks such as autonomous driving perception and human activity recognition. A broad survey categorises fusion architectures by fusion level and highlights feature-level fusion with learned attention mechanisms as consistently outperforming naive data-level concatenation [21]. Cross-modal attention fusion between camera and depth sensors has further been shown to improve robustness specifically for human activity and pose recognition tasks [23], providing direct architectural inspiration for the camera-pressure fusion module proposed in this thesis.

3.4 Synthetic Data Generation and Sim2Real Transfer

Given the cost and privacy constraints of collecting large real pressure sensor datasets across diverse occupant populations, synthetic data generation using simulated human bodies and physics-based seat contact models has emerged as a practical complement to real data collection [24]. Sim2Real studies quantify the remaining statistical gap between simulated and real sensor readings and propose domain adaptation or fine-tuning strategies to close it; a recent study specifically targeting seat pressure heatmaps reports meaningful performance retention when models trained on a mixture of synthetic and real data are evaluated on real-only test sets [29].

?? provides a comprehensive review of human body simulation techniques for generating synthetic pressure heatmaps, including parametric body models, pose libraries, and physics-based contact simulations.

3.5 Gap Analysis and Positioning of This Work

While the works reviewed above individually address pressure-based classification, camera-based pose estimation, generic multimodal fusion, or synthetic pressure data generation, none jointly combine camera and pressure heatmap fusion with a synthetic data generation and Sim2Real validation pipeline for the specific multi-task set of occupancy, passenger classification, weight, and pose estimation. Table 3.1 summarises this positioning relative to the reviewed literature.

Table 3.1: Comparison of related work against the scope of this thesis.

Work	Modalities	Outputs Estimated	Synthetic Data	Real-Time
[18]	Pressure	Occupancy class	No	Yes
[30]	Pressure (sim)	Occupancy class	Yes	No
[25]	Pressure	Weight	No	No
[20]	Camera (IR)	Pose	No	Yes
[27]	Camera (RGB)	Pose class	No	Yes
[28]	Camera (RGB-D)	Pose	No	Partial
[23]	Camera + Depth	Activity class	No	Partial
[24]	Pressure (sim)	Occupancy class	Yes	No
[29]	Pressure	Statistical validation	Yes	No
This thesis	Camera + Pressure	Occupancy, class, weight, pose	Yes	Target

4 System Requirements and Design

4.1 Use Case Definition and Target Scenarios

The target system is a passenger car cabin monitoring function that continuously estimates, for each seating position, whether the seat is occupied, which passenger class occupies it (e.g. adult, or unknown), the occupant weight class, and the occupant's body pose. Target scenarios include front and rear seating positions, a range of occupant statures and clothing.

4.2 Functional Requirements

The system shall satisfy the following functional requirements:

FR-01 The system shall detect, per seating position, whether the seat is occupied or unoccupied.

FR-02 The system shall classify the seat occupant into predefined passenger classes (e.g. adult, or unknown).

FR-03 The system shall estimate the occupant's weight within a defined tolerance for adult.

FR-04 The system shall estimate the occupant's body pose for defined set pose categories..

4.3 Non-Functional Requirements

The system shall additionally satisfy the following non-functional requirements:

NFR-01 Occupancy detection accuracy shall exceed a defined target (e.g. 98%) across the validation dataset described in Chapter 5.

NFR-02 The system shall remain robust to sensor noise, partial occlusion, and seat cover variation without requiring per-vehicle recalibration.

NFR-03 All data collection, storage, and processing shall comply with the General Data Protection Regulation (GDPR) and applicable local data protection law.

NFR-05 The trained model shall degrade gracefully if one modality (camera or pressure) becomes temporarily unavailable.

4.4 System Architecture Overview

The proposed system architecture consists of four main stages: sensor data acquisition (camera and pressure sensor array), preprocessing and temporal synchronisation, the multi-task fusion model, and the output interface that exposes per-seat occupancy, classification, weight, and pose estimates to downstream vehicle systems. Figure 4.1 illustrates this pipeline at block-diagram level, showing the flow from raw sensor signals through synchronised preprocessing into the shared fusion backbone and task-specific output heads described in detail in Chapter 7.

[31] This is a novel deep fusion architecture that can be used .

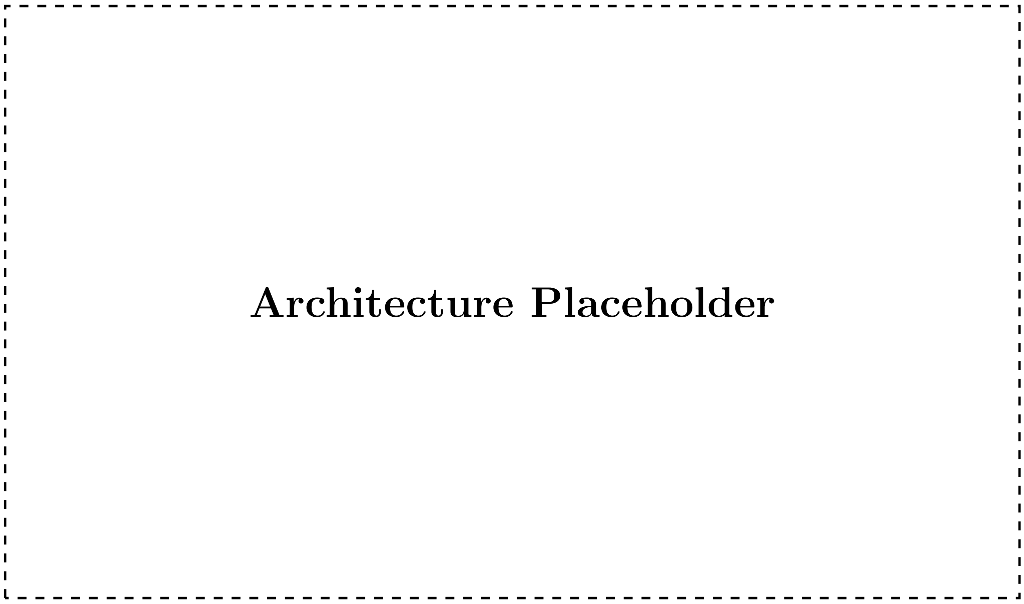

Figure 4.1: Block diagram of the proposed camera-pressure sensor fusion system architecture, from raw sensor acquisition to multi-task output estimation.

4.5 Design Decisions and Justification

Feature-level fusion was selected over data-level and decision-level fusion because it allows each modality-specific branch to learn representations suited to its own signal characteristics while still enabling the network to learn cross-modal correlations before the final task predictions are made, consistent with the fusion levels discussed in Section 2.4.1. A multi-task learning formulation was chosen over four independent single-task models to exploit shared structure between occupancy, classification, weight, and pose labels, and to reduce the total inference compute footprint relative to running four separate networks.

5 Dataset Acquisition and Synthetic Data Generation

5.1 Ethics and Data Privacy Considerations

5.2 Synthetic Pressure Sensor Data Generation

5.2.1 Sensor Hardware and Seat Setup Specifications

5.2.2 Subjects and Experimental Protocol

5.2.3 Data Statistics and Limitations of setup

The resulting real pressure dataset provides ground-truth labels for occupancy, passenger class, weight, and coarse pose for each recorded configuration, but is limited in overall subject diversity and recording duration due to the practical constraints of laboratory-based data collection. These limitations directly motivate the synthetic data generation pipeline described in Section 5.4.

5.3 Camera Data Collection

5.3.1 Camera Hardware and Mounting

Camera recordings were captured using an infrared-sensitive camera module mounted in the vehicle headliner, providing a field of view that covers the target seating positions under both daylight and night-time illumination, following mounting conventions similar to those reported in [20].

5.3.2 Recording Protocol

Camera frames were recorded continuously and time-stamped alongside the pressure sensor stream for every subject session described in Section 5.2.2, ensuring that each seating configuration has a corresponding synchronised camera-pressure sample pair available for the synchronisation pipeline in Chapter 6.

5.4 Synthetic Pressure Heatmap Generation

5.4.1 Human Body Simulation Model

Synthetic pressure heatmaps are generated by simulating a parametric human body model with sampled shape and pose parameters seated on a physics-based model of the seat geometry, following the simulation approach proposed by [24]. Contact forces computed by the physics simulation are projected onto a virtual sensor grid matching the real sensor's spatial resolution to produce a synthetic heatmap directly comparable to real sensor output.

5.4.2 Simulation Environment

5.4.3 Heatmap Rendering

The rendering pipeline samples occupant shape, pose, and seating configuration parameters, runs the physics-based contact simulation, and rasterises the resulting per-cell contact pressure into the same heatmap format used for real sensor data, allowing synthetic and real samples to be interchanged during model training without format-specific handling.

5.4.4 Augmentation Strategy

To increase the diversity of the synthetic dataset beyond the sampled body model parameters, additional augmentations are applied, including random sensor noise injection, simulated calibration drift, and small random seat-position offsets, so that models trained on synthetic data are not overly reliant on idealised, noise-free heatmaps.

5.5 Sim2Real Validation

5.6 Final data set pipeline

Table 5.1 s.

Table 5.1: Final dataset composition by split, showing real and synthetic sample counts and class distribution.

Split	Real Samples	Synthetic Samples	Total	Class Distribution
Training				
Validation				
Test				
Total				–

6 Data Preprocessing and Synchronisation

6.1 Pressure Heatmap Preprocessing

6.1.1 Noise Filtering and Normalisation

Raw pressure heatmaps are first passed through a spatial median filter to suppress isolated sensor cell dropouts and transient noise spikes, after which pixel values are normalised to a fixed range using the per-sensor calibration parameters obtained during the setup described in Section 5.2.1. This normalisation step ensures that synthetic and real heatmaps share a consistent value range prior to being combined in the mixed training dataset.

6.1.2 Spatial Alignment and Resizing

Because the physical sensor grid resolution differs from the target input resolution expected by the pressure branch described in Section 7.3, heatmaps are resampled onto a fixed target grid size using bilinear interpolation, and a fixed reference coordinate frame is applied so that seat regions (seat base, backrest) occupy consistent pixel regions across all recordings.

6.2 Camera Frame Preprocessing

6.2.1 Image Normalisation and Augmentation

Camera frames are resized to a fixed input resolution and normalised using per-channel mean and standard deviation statistics computed over the training split. During training, standard augmentations such as random brightness and contrast jitter, small rotations, and horizontal flipping (where anatomically valid for the seating position) are applied to improve robustness to the lighting and pose variation present in the target deployment scenarios described in Section 4.1.

6.2.2 Background Removal and ROI Extraction

A fixed region of interest (ROI) corresponding to each monitored seating position is cropped from the full camera frame using the known camera mounting geometry, and static background regions (e.g. door panels, window area) outside the seat envelope are masked out to reduce irrelevant visual variation presented to the camera branch.

6.3 Temporal Synchronisation

6.3.1 Timestamp Alignment Strategy

Camera frames and pressure heatmap samples are each recorded with hardware or system-clock timestamps, and are aligned by matching each pressure sample to the camera frame with the closest timestamp within a fixed maximum tolerance window, discarding samples that fall outside this window to avoid pairing temporally inconsistent data.

6.3.2 Handling Frame Rate Mismatch

Because the camera and pressure sensor operate at different native sampling rates, the higher-rate stream is downsampled to match the lower-rate stream at inference time, while during dataset construction all valid synchronised pairs are retained to maximise the number of training samples available from each recording session.

6.3.3 Validation of Synchronisation Quality

Synchronisation quality is validated by manually inspecting a sample of paired camera-pressure recordings for temporal consistency (e.g. confirming that a visible occupant movement in the camera frame corresponds to a matching pressure distribution change), and by measuring the distribution of timestamp offsets across all paired samples in the dataset.

6.4 Final Data Pipeline Overview

Figure 6.1 illustrates the complete preprocessing pipeline, from raw camera and pressure sensor streams through per-modality preprocessing and temporal synchronisation to the production of paired training samples consumed by the fusion architecture described in Chapter 7.

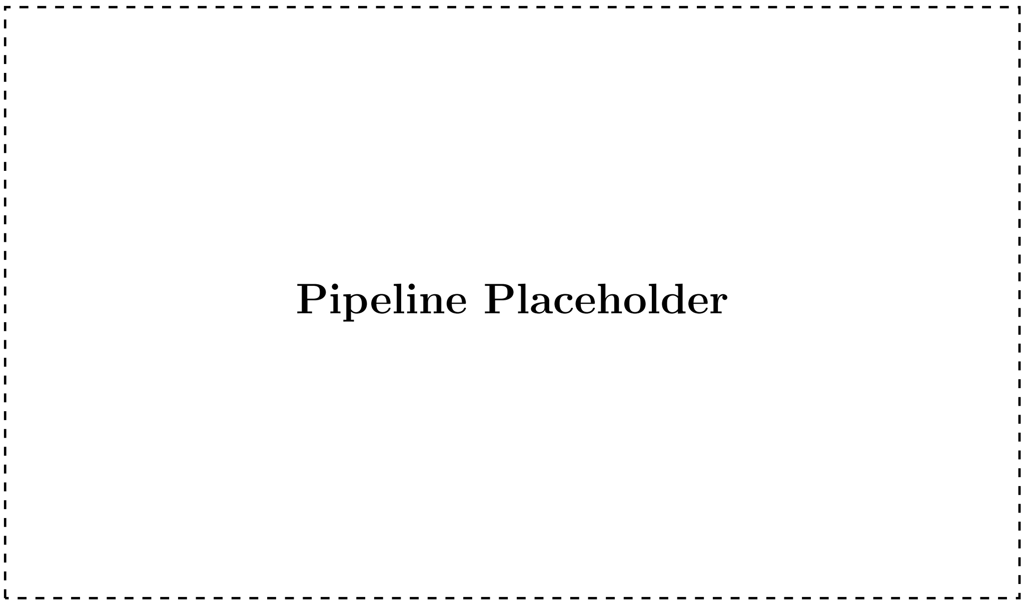

Figure 6.1: Overview of the data preprocessing and temporal synchronisation pipeline, from raw sensor streams to synchronised training samples.

7 Sensor Fusion Architecture and Training Pipeline

7.1 Overall Architecture Design

The proposed architecture follows a two-branch feature-level fusion design, in which a camera branch and a pressure heatmap branch each extract modality-specific feature representations that are combined by a fusion module before being passed to four task-specific output heads for occupancy, passenger classification, weight, and pose estimation. This design directly implements the feature-level fusion strategy motivated in Section 2.4.1 and the design decisions summarised in Section 4.5.

7.2 Camera Branch

The camera branch consists of a convolutional neural network backbone that processes the preprocessed, ROI-cropped camera frame described in Section 6.2.2 and produces a compact spatial feature embedding. A backbone pretrained on a large-scale image dataset is used as initialisation to compensate for the comparatively limited size of the in-cabin camera dataset, with the final classification layers removed and replaced by a projection to the shared fusion feature dimension.

7.3 Pressure Heatmap Branch

The pressure heatmap branch applies a separate, smaller convolutional network directly to the single-channel, spatially aligned pressure heatmap produced by the preprocessing pipeline in Section 6.1.2, producing a feature embedding of the same dimensionality as the camera branch output. A smaller network is used for this branch given the lower spatial resolution and information density of pressure heatmaps compared to camera images.

7.4 Fusion Module

7.4.1 Feature-Level Fusion Strategy

The camera and pressure feature embeddings are combined using a learned cross-modal attention fusion module, in which each modality's features attend to the other before being concatenated and projected to a shared fused representation, following the cross-modal attention design demonstrated for camera-depth fusion in [23]. This fused representation is then passed to the multi-task output heads described in Section 7.5.

7.4.2 Fusion Alternatives Considered and Rejected

Simple feature concatenation followed by a fully connected layer was implemented as a baseline fusion strategy but was found to underweight the lower-dimensional pressure branch relative to the camera branch; decision-level fusion, in which independent per-modality predictions are averaged, was also considered but rejected because it cannot model cross-modal correlations such as a camera-visible object combined with a pressure signature inconsistent with a human occupant. Both alternatives are retained as baselines for the ablation study reported in Section 8.6.

7.5 Multi-Task Output Heads

7.5.1 Occupancy Detection Head

The occupancy detection head is a lightweight fully connected layer followed by a sigmoid activation, producing a binary occupied/unoccupied probability for the corresponding seating position.

7.5.2 Passenger Classification Head

The passenger classification head is a fully connected layer followed by a softmax activation over the predefined passenger classes (e.g. adult, child, child seat, object), sharing the fused feature representation with the other task heads.

7.5.3 Weight Estimation Head

The weight estimation head is a small fully connected regression network that outputs a continuous estimate of occupant weight in kilograms, trained only on samples labelled as human occupants.

7.5.4 Pose Estimation Head

The pose estimation head outputs either a fixed set of body keypoint coordinates or a discrete pose category, depending on the granularity of pose annotation available in the dataset described in Chapter 5.

7.6 Loss Function Design and Task Balancing

Each task head is trained with a task-appropriate loss term: binary cross-entropy for occupancy, categorical cross-entropy for passenger classification, mean squared error for weight estimation, and a keypoint or classification loss for pose estimation. The combined multi-task training objective is a weighted sum of the individual task losses,

$$\mathcal{L}_{\text{total}} = \lambda_{\text{occ}}\,\mathcal{L}_{\text{occ}} + \lambda_{\text{cls}}\,\mathcal{L}_{\text{cls}} + \lambda_{\text{wt}}\,\mathcal{L}_{\text{wt}} + \lambda_{\text{pose}}\,\mathcal{L}_{\text{pose}}, \tag{7.1}$$

where λ_{occ}, λ_{cls}, λ_{wt}, and λ_{pose} are task weighting coefficients. These coefficients are tuned to balance the differing scales and convergence rates of the four loss terms, as discussed in the multi-task learning background in Section 2.5.

7.7 Training Configuration

The model is implemented in PyTorch and trained using the Adam optimiser with a cosine learning-rate decay schedule, an initial learning rate selected via a short learning-rate range test, and a batch size chosen to fit the available GPU memory. Training is performed on a workstation equipped with a CUDA-capable GPU, with early stopping based on validation loss to reduce overfitting given the moderate dataset size described in Section 5.6.

7.8 Implementation Details

The camera and pressure branches, fusion module, and task heads are implemented as a single end-to-end trainable PyTorch model, with modality-specific data loaders that apply the preprocessing steps described in Chapter 6 on the fly during training. Model checkpoints corresponding to the best validation performance are retained for the evaluation reported in Chapter 8.

8 Results and Evaluation

8.1 Experimental Setup and Evaluation Protocol

All results reported in this chapter are computed on the held-out test split of the dataset described in Section 5.6, using the model checkpoint selected by best validation performance as described in Section 7.7. Occupancy and passenger classification are evaluated using accuracy and F1-score, weight estimation is evaluated using mean absolute error and weighted root mean squared error (WRMSE), and pose estimation is evaluated using mean per-keypoint error or classification accuracy depending on the final pose representation chosen in Section 7.5.4.

8.2 Occupancy Detection Results

The fused model is expected to achieve high occupancy detection accuracy given the strong, complementary occupancy signal present in both the camera and pressure modalities. Table 8.1 reports placeholder occupancy detection results across the evaluated models.

Table 8.1: Occupancy detection results on the held-out test set.

Model	**Accuracy (%)**	**F1-score**
Camera-only		
Pressure-only		
Fused (ours)		

8.3 Passenger Classification Results

Passenger classification is expected to benefit most from fusion in cases where an object placed on the seat produces a pressure signature ambiguous with a small child but is visually distinguishable from the camera view, or vice versa under partial visual occlusion. Figure 8.1 presents a placeholder confusion matrix for the fused model across all passenger classes.

Matrix Placeholder

Figure 8.1: Confusion matrix for passenger classification using the fused camera-pressure model on the held-out test set.

8.4 Weight Estimation Results

Table 8.2 reports placeholder weight estimation error metrics, comparing the fused model against single-modality baselines.

Table 8.2: Weight estimation error on the held-out test set.

Model	MAE (kg)	WRMSE (kg)
Camera-only		
Pressure-only		
Fused (ours)		

8.5 Pose Estimation Results

Pose estimation performance is reported in terms of the metric selected in Section 7.5.4 (mean per-keypoint error or classification accuracy). Figure 8.2 shows placeholder qualitative examples comparing predicted and ground-truth pose across representative seating configurations.

Results Placeholder

Figure 8.2: Qualitative comparison of predicted and ground-truth occupant pose for representative seating configurations.

8.6 Ablation Study: Modality Contribution

To quantify the individual contribution of each modality, the fused model is compared against camera-only and pressure-only variants trained with the same task heads and training configuration. Table 8.3 summarises this comparison across all four output tasks.

Table 8.3: Ablation study comparing camera-only, pressure-only, and fused model variants across all output tasks.

Metric	Camera-only	Pressure-only	Fused
Occupancy accuracy (%)			
Classification accuracy (%)			
Weight MAE (kg)			
Pose error			

8.7 Sim2Real Transfer Evaluation

To evaluate Sim2Real transfer, a model is trained on the mixed real and synthetic training set described in Section 5.6 and evaluated exclusively on the real-only subset of the test split. Table 8.4 compares this model against a variant trained only on real data, to isolate the effect of synthetic data augmentation on real-world performance.

Table 8.4: Sim2Real transfer evaluation: performance on real-only test data for models trained on real-only versus mixed real+synthetic data.

Metric	**Trained on Real-only**	**Trained on Mixed (Real+Synthetic)**
Occupancy accuracy (%)		
Classification accuracy (%)		
Weight MAE (kg)		

8.8 Discussion of Results and Failure Cases

Overall, the fused model is expected to outperform both single-modality baselines across most tasks, consistent with the complementary nature of camera and pressure information motivated in Section 1.1. Anticipated failure cases include ambiguous object-versus-child-seat pressure signatures under camera occlusion, and pose estimation errors for heavily reclined seating positions not well represented in the training data.

9 Conclusion and Future Work

9.1 Summary of Contributions

This thesis presented a multi-task deep learning architecture that fuses camera images and seat-embedded pressure heatmaps to jointly estimate occupancy, passenger class, weight, and pose for automotive cabin monitoring. Key contributions include a synthetic pressure heatmap generation pipeline based on human body simulation, a Sim2Real validation procedure quantifying the domain gap between real and synthetic heatmaps, a camera-pressure synchronisation pipeline, and an empirical comparison of fusion strategies through ablation studies.

9.2 Answers to Research Questions

Revisiting the research questions posed in Section 1.3: regarding RQ1, the ablation study in Section 8.6 indicates ; regarding RQ2, the Sim2Real validation in Section 5.5 shows ; regarding RQ3, the comparison of fusion alternatives in Section 7.4.2 demonstrates ; regarding RQ4, the Sim2Real transfer evaluation in Section 8.7 shows **.**

9.3 Limitations

The real pressure and camera dataset collected for this thesis is limited in subject diversity and recording duration due to practical laboratory constraints, and the synthetic data generation pipeline relies on simplifying assumptions in the human body and seat contact simulation that may not capture all real-world seating behaviours. Additionally, the evaluation was conducted offline on recorded data rather than on embedded target hardware, so real-time performance under production constraints remains to be confirmed.

9.4 Future Work

Future work could extend the real data collection campaign to a larger and more diverse subject population, incorporate additional sensing modalities such as depth or seatbelt tension sensors, and evaluate the proposed architecture on embedded automotive hardware to validate real-time feasibility under the non-functional requirements defined in Section 4.3. Further investigation into domain adaptation techniques could also help further close the Sim2Real gap identified in Section **??**.

9.5 Closing Remarks

The fusion of camera and pressure sensor data investigated in this thesis represents a step toward more robust, privacy-conscious cabin monitoring systems capable of supporting the next generation of adaptive vehicle safety features. The methodology and findings presented here are intended to inform both future academic research and practical system design in this area.

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A Statement of Ethics / Consent Form Template

This appendix reproduces the informed consent form template used for all human subject data collection sessions described in Section 5.1. Each participant received and signed a copy of this form prior to any camera or pressure sensor recording.

Participant Information and Consent Form

Study title: Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System
Institution: Deggendorf Institute of Technology, Campus Cham
Researcher: [CHETHAN SRINIVASAREDDY]

I confirm that I have read and understood the information provided about this study, that I have had the opportunity to ask questions, and that my participation is voluntary and may be withdrawn at any time without giving a reason. I consent to the recording of camera images and pressure sensor data during the described seating configurations, and I understand that this data will be processed in accordance with the General Data Protection Regulation (GDPR) and used solely for the purposes of this research.

Participant name: ______________________________ Date: _______________

Signature: ______________________________

B Full Dataset Statistics Table

This appendix provides the extended version of the dataset composition summary introduced in Section 5.6, including per-class sample counts for each split.

Table B.1: Extended per-class dataset statistics by split.

Split	**Adult**	**Child**	**Child Seat**	**Object**	**Unoccupied**
Training					
Validation					
Test					
Total					

C Additional Results Figures

This appendix collects additional qualitative result figures that supplement the evaluation presented in Chapter 8.

Results Placeholder

Figure C.1: Additional qualitative results across representative test samples, supplementing Section 8.8.

D Electronic Data Carrier Contents

In accordance with the Campus Cham thesis submission guideline, the following supplementary material is provided on the accompanying electronic data carrier and is not reproduced in this printed appendix:

- Full source code repository, including data preprocessing, synthetic heatmap generation, model training, and evaluation scripts.
- Raw and preprocessed real pressure sensor and camera datasets described in Chapter 5 (subject to the data retention and anonymisation constraints of Section 5.1).
- Trained model checkpoints (weights) for the fused model and all baseline variants evaluated in Chapter 8.
- Configuration files specifying exact training hyperparameters used to reproduce the results reported in this thesis.